

THE THREE GEOMETRIES OF A LEARNING PROBLEM

1. DOMAIN

Where the data live, Ω



Grid

Sphere

Graph

Grid, sphere, point cloud, graph
Translation, rotation, permutation

2. SIGNAL

What lives on it

Scalar Vector Tensor



$$s \mapsto s$$

Invariant



$$\mathbf{v} \mapsto R\mathbf{v}$$

Equivariant



$$\mathbf{T} \mapsto R\mathbf{T}R^\top$$

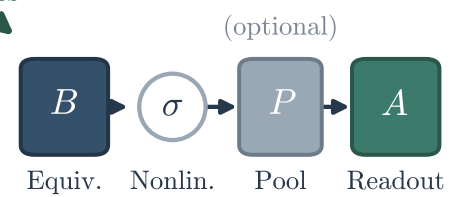
Equivariant

A field on Ω , with its symmetry
(Here R is a rotation)

Implies

3. MODEL

What we build



$$f = A \circ (P) \circ \sigma \circ B \circ \dots \circ B$$

The geometric blueprint
Its symmetry $\subseteq$ the domain's

GENERAL

GALAXY

Image on a grid



Translations $T(2)$,
90° rotations (D_4)

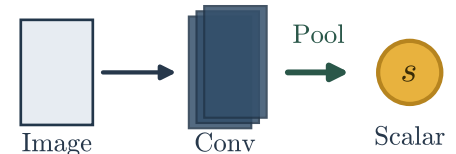
Example inference task:
Time since last merger



Rotate by any θ : Same answer
 $\Rightarrow$ invariant under $SO(2)$

Implies

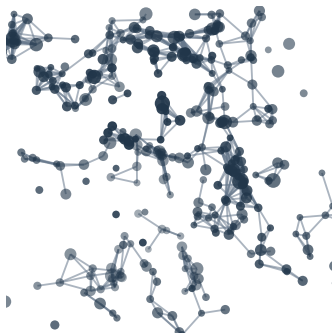
CNN



Equivariant convs,
then global pool to a scalar

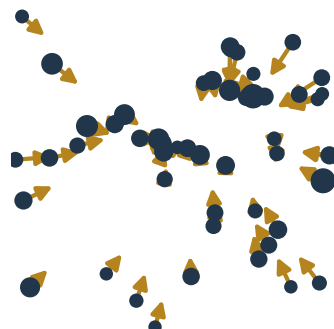
HALOS

Halo point cloud



Permutation Σ_n ;
Linking length $\Rightarrow$ locality

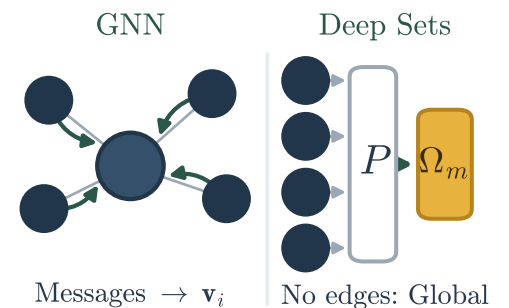
Example inference task:
Velocity vectors, or Λ CDM



Vector: $O(3)$ -equivariant
 Λ CDM: $E(3)$ -invariant

Implies

GNN / Deep Sets



Use GNN for local interactions;
Deep Sets for global param invariance.

Left to right: The domain and the signal imply the model.